Exercise 5

Exercise 5.1 – SD card

Read and/or write from/to a SD Card. The exercise can be approached by using:

- the OpenLog, which provides a UART interface to the SD card
- the SPI interface directly on the SD card adaptor (Optional).





OpenLog Datalogger *microSD Card Module*

Introduction to OpenLogger: The OpenLog Data Logger is a simple-to-use, open-source solution for logging serial data from your projects. The OpenLog provides a simple serial interface to log data from a project to a microSD card. (Excerpts from https://learn.sparkfun.com/tutorials/openlog-hookup-guide)

OpenLog Power Ratings	
VCC Input	3.3V-12V (Recommended 3.3V-5V)
RXI Input	2.0V-3.8V
TXO Output	3.3V
Idle Current Draw	~2mA-5mA (w/out microSD card) ~5mA-6mA (w/ microSD card)
Active Writing Current Draw	~20-23mA (w/ microSD card)

The OpenLog's current draw is about 20mA to 23mA when writing to a microSD. Depending on the size of the microSD card and its manufacturer, the active current draw can vary when the OpenLog is writing to the memory card. Increasing the baud rate will also pull more current.

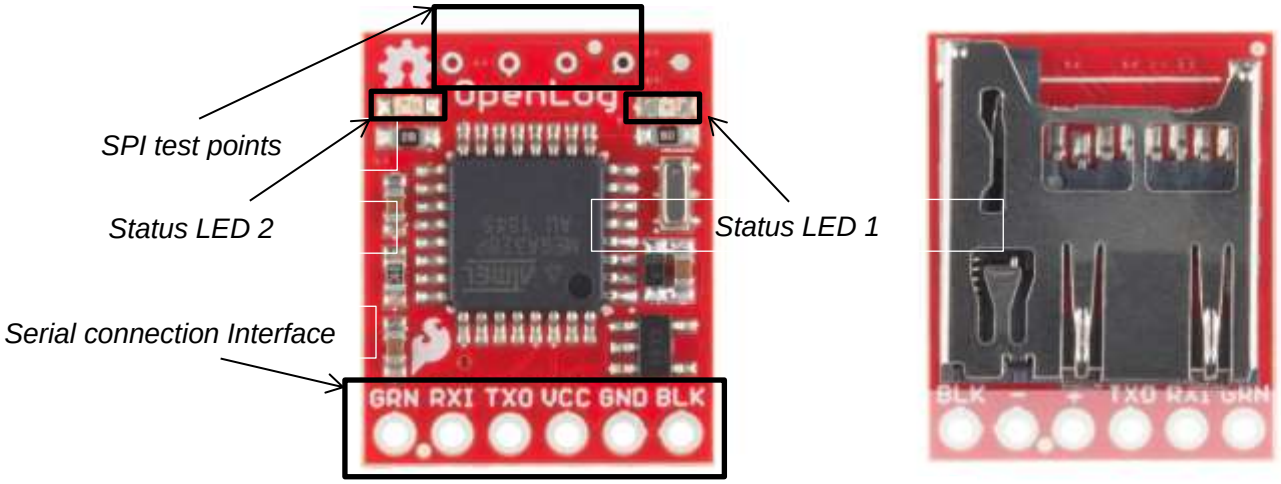
Microcontroller on the OpenLog: The OpenLog runs off of an onboard ATmega328, running at 16MHz thanks to the onboard crystal. The ATmega328 has the Optiboot bootloader loaded on it, but don't need to change this configuration.

Interface of the OpenLog: The primary interface with the OpenLog is the header pins on the board edge. This header connections allows the microcontroller to send the data over a serial connection to the OpenLog. There are also four SPI test points broken out on the opposite end of the board. You can use these to reprogram the bootloader on the ATmega328, but please don't change the configuration of the OpenLog. The final interface for communicating with the OpenLog is the microSD card itself. Not only is this where the data is stored by the OpenLog, but you can also update the OpenLog's configuration via the config.txt file on the microSD card. All data logged by the OpenLog is stored on the microSD card. The OpenLog works with microSD cards that involve the following features: 64MB to 32GB / FAT16 or FAT32

Status LED: There are two status LEDs on the OpenLog to help you with troubleshooting.

STAT1 - This blue indicator LED is attached to Arduino D5 (ATmega328 PD5) and toggles on/off when a new character is received. This LED blinks when Serial communication is functioning.

STAT2 - This green LED is connected to Arduino D13 (SPI Serial Clock Line/ ATmega328 PB5). This LED only blinks when the SPI interface is active. You will see it flash when the OpenLog records 512 bytes to the microSD card.



OpenLog Datalogger front (left) and back (right) sides

OpenLog Hardware Connection:

While interfacing with the OpenLog over a serial connection is important for reprogramming or debugging, the place where OpenLog shines is in an embedded project. This general circuit is how we recommend you hook up your OpenLog to a microcontroller (our ARM STM32) that will write serial data out to the OpenLog.

First you will need to develop the code to your STM32 you intend to run. Please check out the some Sketches for some example code that you can use as inspiration. Make sure to connect the pins labeled BLK and GRN on both the STM32 and OpenLog. Use jumper wires and make the connections on the table on the right.

OpenLog		**STM32**
Label on Board	*Function*	*Connection*
BLK	Connected to GND	GND
GND	GND	GND
VCC	VCC	VCC
TX0	TX	Serial1_RX
RXI	RX	Serial1_TX
GRN	DTR	Serial1_RTS

Exercise 4.2 – Choose an Interface

See different components on the Course folder. As starting point you shall use the OpenLog, but you instead can use the SPI interface/adaptor for accessing the SD card.

Report

Write a report of the exercise. The report should be about 5 pages long including a description, a chart, a table of used ports, documentation of results (eg. screendumps) and a well commended featured source code including header file.

Participants on a team draw up a single report with a cover, which show participants names, student number and group number. It should not be specified in the text, who wrote what. The report is uploaded preferably as PDF file in the course Assignment. The team will have an annotated report back via e -mail.

The report should typically include most of these standard points:

Title

Abstract Subject , methods, results.

Introduction general description, no details.

Problem definition What is it about.

Problem limitation What you cannot with this method or this hardware.

Theory How do the various components and methods.

Hardware Diagram and table of used ports.

Software Function Description (no source code here).

Test method/software What worked, what did not and why.

Conclusion Must correspond to the problem formulation.

Appendix Source code, diagrams, possibly. Print drawings,

Data Sheets (only important pages), references and links.

Zip file with source code

A separate file with the c and h files shall be also delivered.

If other code has been used (matlab, python or other), this shall be also be delivered.

If data (from modules or similar) has been used, this shall be also be delivered.